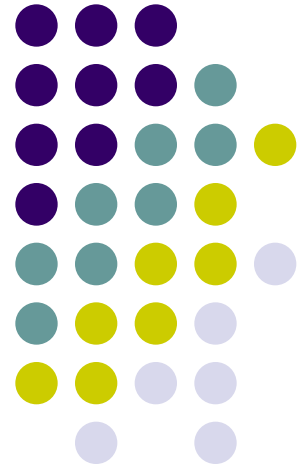


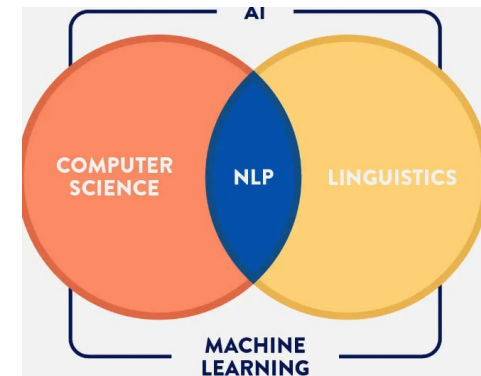
Natural Language Processing

CPSC 488

An Overview of Natural Language Processing (NLP)



Natural Language Processing

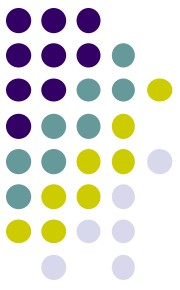


- **What is NLP?**

- A field of AI focusing on enabling computers to process, interpret and understand, and generate natural language
- What is natural language?
 - Any language that occurs naturally in a human community

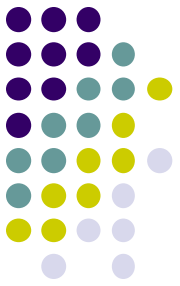
- **What does an NLP system need to achieve?**

- A natural language consists of many levels of structure or rules with words.
- Humans fluently integrate all of these in understanding and producing language.
- Ideally, so would a computer!

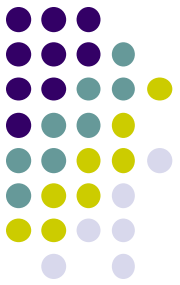


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- **Acknowledgement** to the following scholars for many PPT slides:
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 - Phili Blunsom@Oxford, P. Koehn@Johns Hopkins, A. Lascarides and S. Cohen@UnivEdinburgh
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- J. Eisenstein, **Natural Language Processing**, Draft, 2018.
- D. Rao and B. McMahan, **Natural Language Processing with PyTorch: Build Intelligent Language Applications using Deep Learning**, O'Reilly, 2019.
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Linguistics Terminologies



Linguistic and NLP Terms

- **Character, string, token, word, text**

- Word vs token
 - **Word**: a vocabulary, a unit of **meaning** in a language
 - **Token**: a specific instance of a word or symbol in text

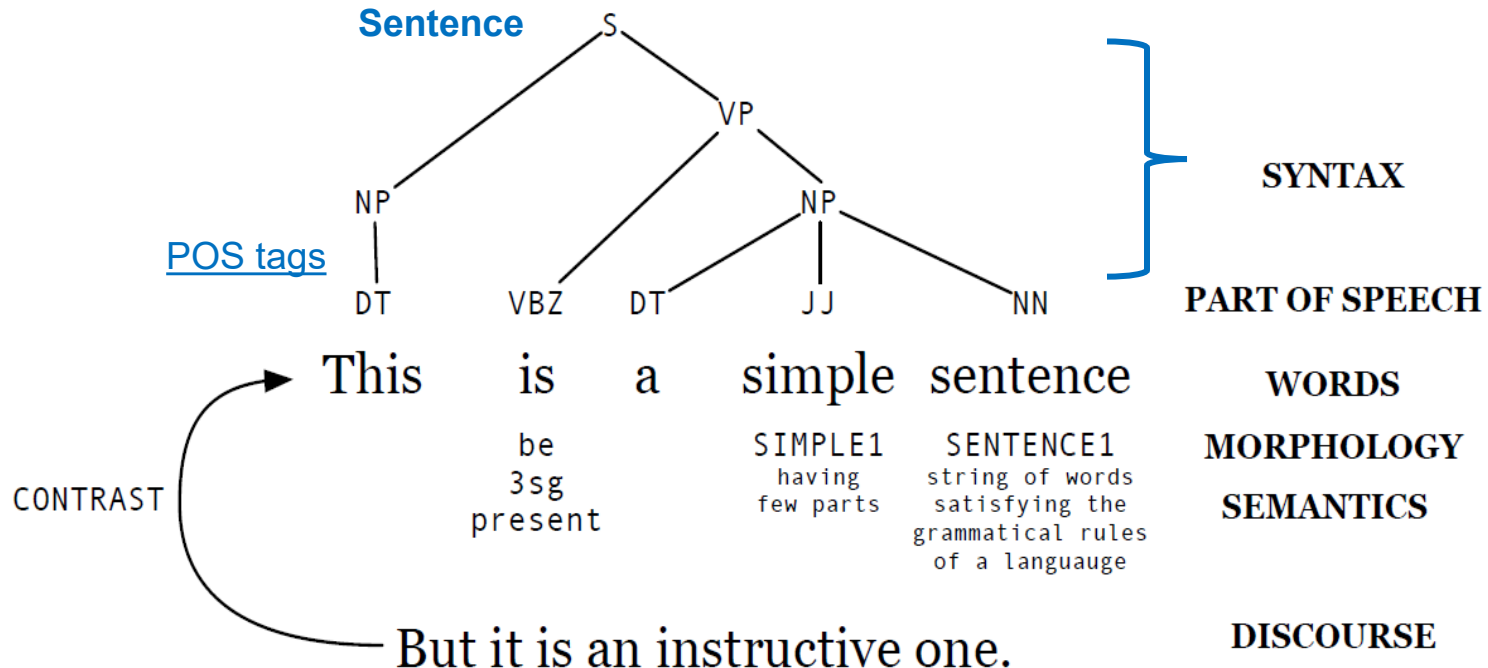
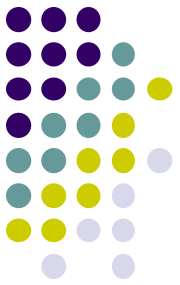
- **Morphology**

- A study of words, how they are formed, their relationships with other words such as:
 - **Wordform**: the full inflected surface form, e.g., cat and cats are different wordforms.
 - **Morphemes**: the smallest units of language (cannot be divided without changing it meaning), e.g., “dog”, “happy”, “run”, “ly” are morphemes but “friendly” is not a morpheme.
 - **Subword** is parts of a word such as prefixes, suffixes, stems. E.g., for “Unfriendly”, “friend” is stem; “Un-” is prefix; and “-ly” is suffix.
 - Morphemes correspond to sound: “**Phonemes**” in “**Phonology**” (the smallest units of sound in a language)

- **Sentence**

- A grammatical unit composed of one or more clauses, expressing a complete thought

A Structure of a Sentence



- **Part-Of-Speech (POS) tags**

- **Labels** used to indicate Part-Of-Speech and grammatical categories
 - S (sentence), N (common noun), NP(noun phrase), NN (noun, singular or mass), V (verb), VP (verb phrase), PP (Prepositional phrase) DT (determiner), VBZ (verb in the third person singular present tense), JJ (adjective), etc. (in **tagset** from *the Penn Treebank*)

- **Discourse** – the expression of thought, conversation



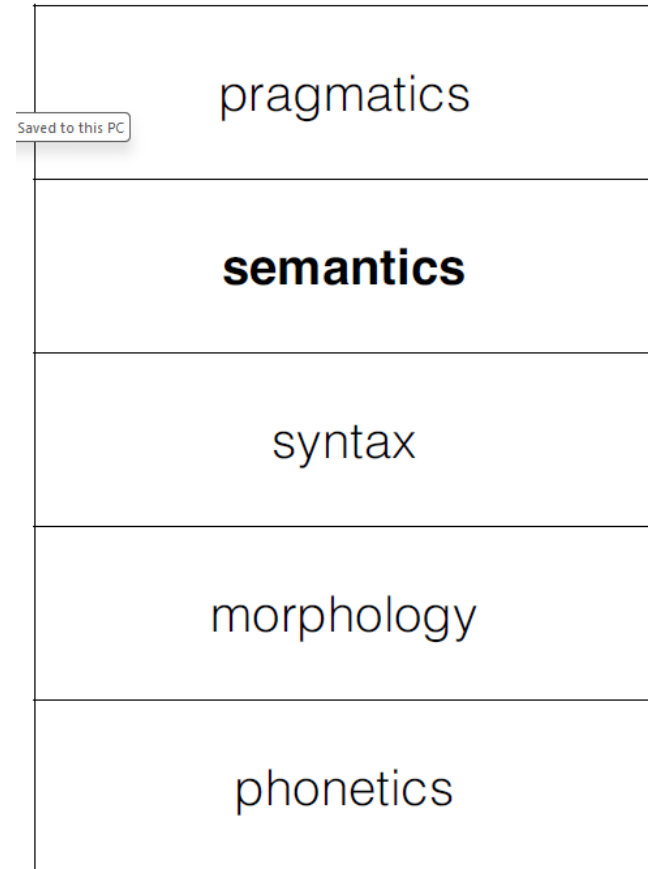
Linguistic and NLP Terms

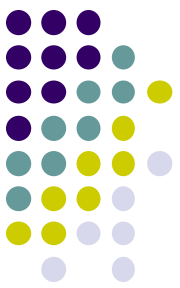
- **Stem** – core form of a word (**form dependent**)
 - Used in *morphological analysis* to identify the base form of a word
 - The stem for “**running**” and “**ran**” is “**run**” (The core meaning remains unchanged.)
 - The stem for “**feet**” is “**feet**”. (irregular words, e.g., feets is possible)
- **Lemma** – canonical or dictionary form of a word (**form independent**)
 - The lemma for “**running**” and “**ran**” is “**run**”.
 - The lemma for “**feet**” is “**foot**”. The lemma for “**men**” is “**man**”.
 - The **stem** for “**flies**” is “**fli**” and lemma is “**fly**”.
- **Syntax** – word order
- **Grammar** – structural rules of a language
- **Semantics** – general meaning of words in language
 - Lexical words: content words with **concrete meaning** in sentences
 - Denotation of a word: **literal meaning** of a word (in dictionary, a part of lexical meaning)
 - Connotation of a word: possible **meanings associated** with a word
 - Functional words: words that serve a **grammatical function** in a sentence
 - For example, “**a**”, “**the**”, “**to**” are functional words.
- **Pragmatics** – meanings of words in **context**
 - A study of how people use language in context, “**what speakers mean**”



A Linguistic Hierarchy

- Linguistic hierarchy
 - The structured arrangement of language element
- Pragmatics
 - Concerns how speakers imply meaning and how listeners infer it, taking into account factors like the speaker's intention, the context of the interaction, and social conventions.





Text Data: Document and Corpus

- **Document**

- A collection of text as a unit for processing

- **Corpus** (plural corpora or corpuses)

- A large or complete collection of documents
 - Example corpora: **Brown corpus** (1961), **WSJ** (1989), **Switchboard** (1990-91)
- **Example corpora** (released by the Linguistic Data Consortium)

Brown: 500 texts, 1M words in 15 genres. POS-tagged. **SemCor** subset (234K words) labelled with WordNet word senses.

WSJ: 6 years of *Wall Street Journal*; subsequently used to create Penn Treebank, PropBank, and more! Translated into Czech for the **Prague Czech-English Dependency Treebank**.

ECI: European Corpus Initiative, multilingual.

BNC: 100M words; balanced selection of written and spoken genres.

Redwoods: Treebank aligned to wide-coverage grammar; several genres.

Gigaword: 1B words of news text.

AMI: Multimedia (video, audio, synchronised transcripts).

Google Books N-grams: 5M books, 500B words (361B English).

Flickr 8K: images with NL captions

English Visual Genome: Images, bounding boxes $\Rightarrow$ NL descriptions

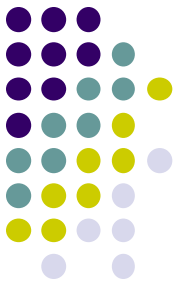
Several common markup formats

for structuring linguistic data:

XML, JSON, CoNLL-style (one token per line)

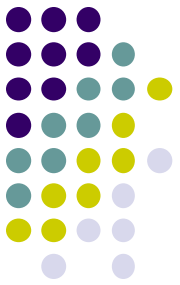
Some datasets such as **WordNet** and **PropBank**, use custom file formats.

NLTK provides friendly Python APIs for reading many corpora.



Corpora Vary along Dimension

- **Text data are produced by**
 - Specific writers at specific times
 - Influenced by the author's demographics (e.g., gender, ethnicity, age)
 - Language variety
 - e.g., >7000 languages in the world)
 - Function
 - e.g., genre like newswire, fiction, scientific articles, Wikipedia)
- **Corpus datasheets include:**
 - **Motivation:** Purpose, collector, and funding
 - **Situation:** Context in which the text was written
 - **Collection process:** Sampling method, consent, preprocessing, annotation
 - **Metadata:** Author info, data, topic, and publication details



Corpus Size

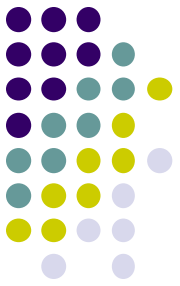
- **Corpus size**

- Number of words in a corpus

- **Heaps' (Herdan's) Law:** $|V| = kN^\beta$

- N = number of tokens; V = vocabulary (unique tokens), $|V|$ is size of vocabulary; $.67 < \beta < .75$
 - Every occurrence of a word is counted as a token separately.
 - K is a scaling constant that depends on the corpus, typically, $30 < k < 100$
- Vocabulary size and number of tokens
 - Vocabulary size grows with β square root of the number of word tokens

Corpus	Tokens = N	Types = $ V $
Switchboard phone conversations	2.4 million	20 thousand
Shakespeare	884,000	31 thousand
COCA (American English)	440 million	2 million
Google N-grams	1 trillion	13+ million

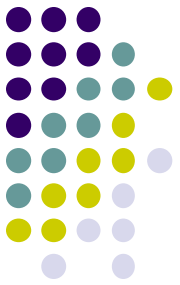


NLP Tasks



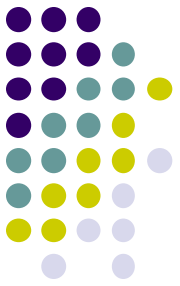
Foundational NLP Tasks

- **Text processing**
 - Tokenization, word normalization, syntactic parsing, vectorization of text data
- **Text classification, sentiment analysis, and clustering**
- **Language modeling**
- **Sequence labeling and modeling**
 - Part-Of-Speech (POS), Named Entity Recognition (NER)
 - Learning sequential data
- **Semantics and discourse**
 - Semantic analysis, Topic modeling
 - Semantic role labeling
 - Identifying the roles of entities to extract the underlying meaning of a sentence
 - Coreference resolution
 - Identifying all expressions that refer to the same entity for question answering
- **Information extraction**
- **Speech recognition and synthesis**
 - Text-to-speech, speech-to-text, and speech synthesis (generation)

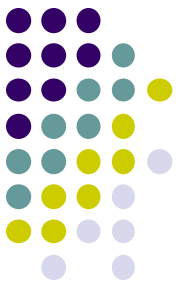


NLP Applications

- Text classification:
 - Sentiment analysis, spam detection, recommendation systems
- Grammar and spelling correction
- Information retrieval, information extraction, summarization
- Machine translation
- Reasoning, language understanding, and language generation
- Speech recognition and voice synthesis
- Communication:
 - Question answering, chatbots, dialogue systems, prompting
- Generative AI:
 - Language generation, code generation, image generation, text-image synthesis
- Multimodal learning
- Intelligent agents



Challenges in NLP

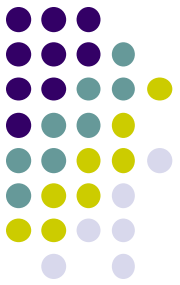


Challenge: Sparse Data

Frequency of words in the English Europarl corpus out of *24 millions word token*

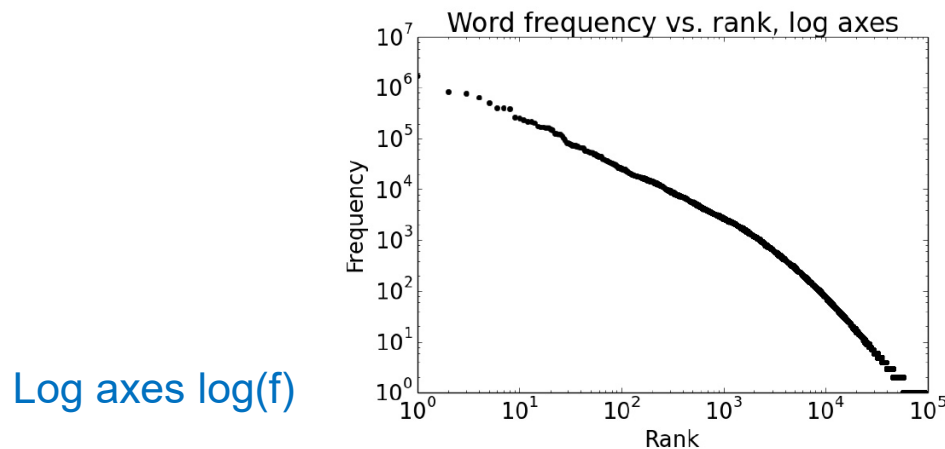
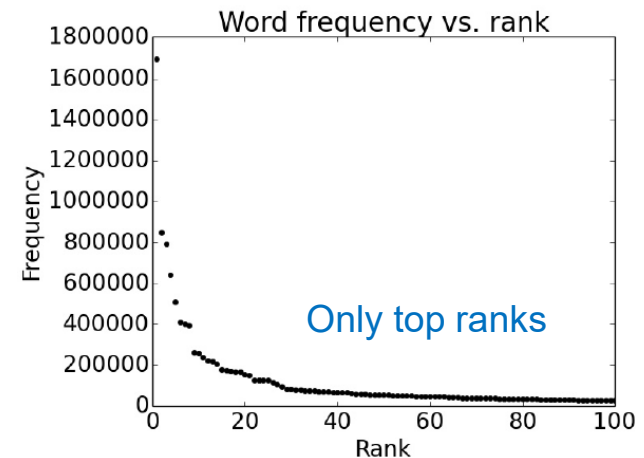
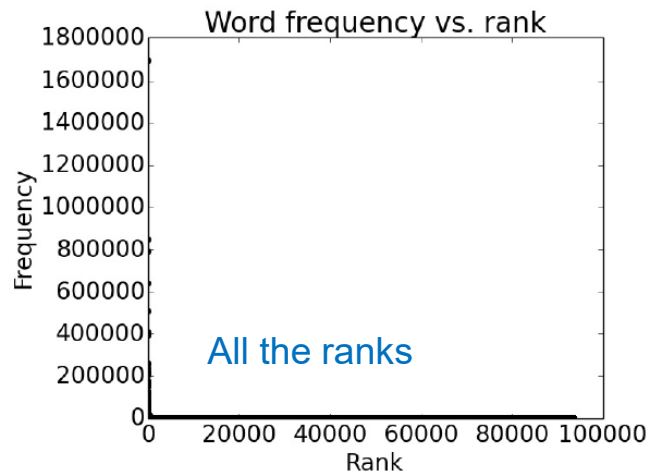
any word		nouns	
Frequency	Type	Frequency	Type
1,698,599	the	124,598	European
849,256	of	104,325	Mr
793,731	to	92,195	Commission
640,257	and	66,781	President
508,560	in	62,867	Parliament
407,638	that	57,804	Union
400,467	is	53,683	report
394,778	a	53,547	Council
263,040	I	45,842	States

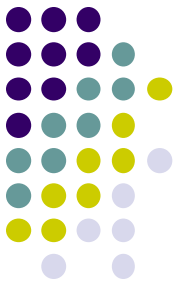
- **Zipf's law: $f \times r \approx k$**
 - f = frequency of a word
 - r = rank of a word frequency
 - k = a constant
- **Word counts from a large text corpus**
 - **36231** of words out of **93638** distinct word types occur **only once**. Refer to Heap's law.
 - What does this tell us?



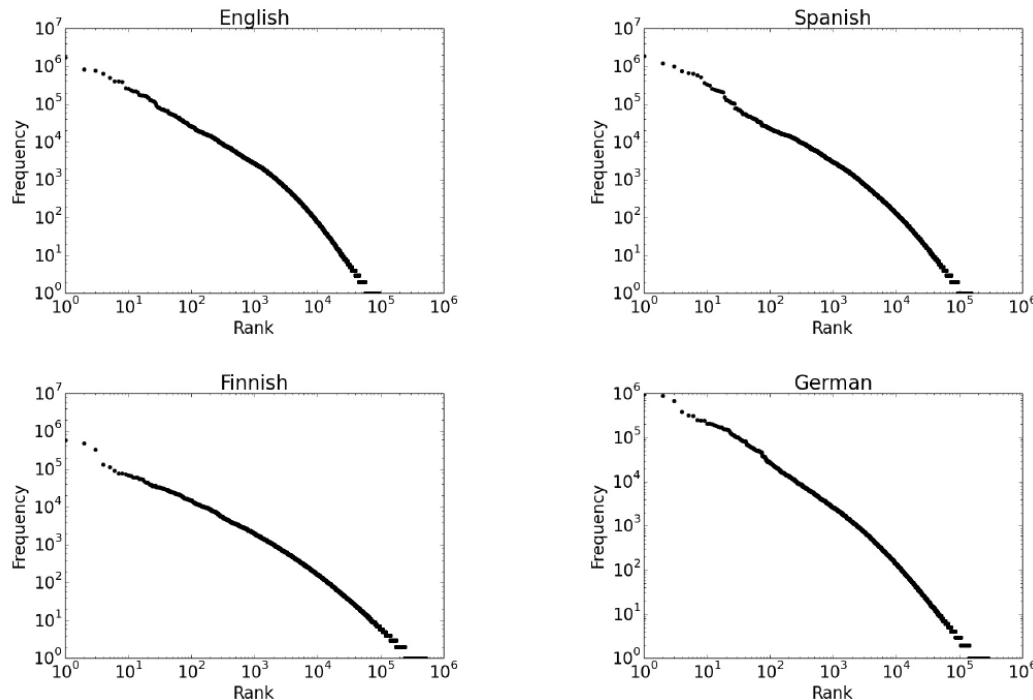
Word Frequency Rank

- What is the frequency of n-th ranked word?





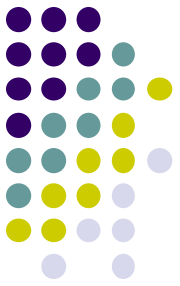
Word Frequency Ranks



● Implications of Zipf's Law

- Regardless of how large our corpus is, there will be a lot of infrequent (or zero-frequent) words. In fact, the same holds for many other levels of linguistic structure (e.g., syntactic rules in a Context-Free Grammar).
 - This means we need to find clever ways to estimate probabilities for things we have rarely or never seen during training.

Challenge: Variations and Expressivity

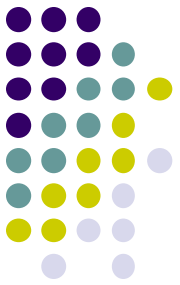


- **Variations of data**

- Suppose we train a part of speech tagger on the Wall Street Journal:
 - Mr./NNP Vinken/NNP is/VBZ chairman/NN of/IN Elsevier/NNP
- What will happen if we try to use this tagger for social media?
 - lkr smh he asked fir yo last name (Twitter example)

- **Expressivity**

- Not only can one form have different meanings (ambiguity), but the same meaning can be expressed with different forms:
 - She gave the book to Tom vs. She gave Tom the book
 - Some kids popped by vs. A few children visited
 - Is that window still open vs. Please close the window



Challenge: Ambiguity

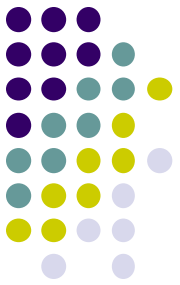
- **Ambiguity at many levels**

- Word sense: bank (finance or river?)
- Part of speech: chair (noun or verb?)
- Syntactic structure: I saw a man with a telescope.
- Quantifier scope: Every child loves some movie.
- Multiple: I saw her duck.
- Reference: John dropped the goblet onto the glass table and it broke.
- Discourse: The meeting is cancelled. Nicholas isn't coming to the office today.

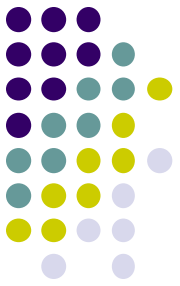
- **Some ways to deal with ambiguity**

- **Rule-based** (non-probabilistic) methods
 - Finite state machines for morphology, parsers for syntax return all possible analyses.
- **Probabilistic models**
 - HMMs for POS tagging and algorithms that returns the best possible analysis
 - Today standard approaches as they are more robust than rule-based methods.

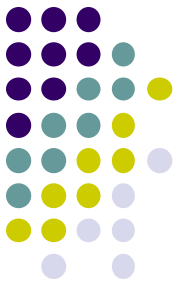
Challenge: Context Dependence



- **Context dependence and unknown knowledge representation**
 - Correct interpretation is context-dependent and often requires world knowledge.
 - Very difficult to capture, since we don't even know how to represent the knowledge a human has/needs:
 - What is the “meaning” of a word or sentence?
 - How do we model context or other general knowledge?
 - Still very hard!
- Remarkably little progress on the Knowledge Representation

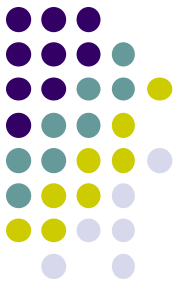


NLP Tools



Some NLP Packages

- WordNet
 - A large lexical database of English words, senses, and their semantic relations
- Natural Language Toolkit (NLTK)
 - Named Entity Recognition (NER), Part-Of-Speech (POS) tagging, Tokenization, Stemming, Semantic reasoning, Text classification
- Apache OpenNLP
 - Tokenization, POS tagging, NER, language detection
- GenSim
 - Semantic similarity
- SpaceCy
 - Multi-trained transformers, Tokenization in >49 languages, Text classification, Sentence segmentation, Lemmatization, POS tagging, NER
- fastText
 - Word embeddings and text classification
- Scikit-learn
 - Neural networks for text preprocessing, POS tagging
- PyTorch-NLP
- TextBlob, AllenNLP, Polyglot



Frameworks, Platforms, and LLMs

- Huggingface
 - Platforms for language models with 20,000 models, >2,000 datasets
- Langchain
 - Language model framework
- Transformer-based Large Language Models (LLMs)
 - BERT, RoBerta
 - GPT, GPT4All, etc.
 - Bard, LaMDA (Google)
 - Gemini
 - Llama (Meta)
 - Dolly
 - Etc.